

What relationship variables predict a more reliable proxy reporter of adolescent wellbeing? A systematic review

Kirstie Northfield^{a,*}, Anthony J. Saliba^a, Keith M. Harris^b

^a School of Psychology, Charles Sturt University, Nathan Cobb Drive, Wagga Wagga, NSW, Australia, 2678

^b Charles Sturt University, 7 Major Innes Dr, Port Macquarie, NSW 2444, Australia

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ABSTRACT

Introduction: Adolescents often face challenges in maintaining their wellbeing during adolescence. While numerous support services are available, their referral systems depend on adolescent self-reports or proxy referrals using ambiguous behavioural cues.

Objectives: This review examines the alignment between proxy reports and self-reports of adolescent wellbeing, aiming to identify predictors of closely matched reports in non-clinical adolescent populations. Further, the review aims to identify variables that predict closely matched reports to determine the characteristics of a more accurate proxy assessor of adolescent wellbeing.

Methods: A systematic literature search was conducted in September 2021 following PRISMA guidelines. Four databases (Scopus, PsycINFO, Medline, and Ebscohost) were searched for studies assessing wellbeing or related constructs in non-clinical adolescents aged 12 to 18. Studies in which there were both self-report and a proxy report, and that assessed concordance between the two reporters were included.

Results: Twenty-one studies met inclusion criteria, showing moderate concordance between parents and adolescents, with less agreement between teachers and adolescents. However, there were few teacher-adolescent studies. The findings indicated that predictors of concordance are more likely to be relationship factors, such as communication, closeness and warmth, rather than age, gender, or proxy role. Patterns emerged that indicate mothers' reports of adolescent wellbeing, as compared to fathers was more closely matched to adolescent self-reports on psychological and emotional domains.

Conclusion: Adolescents' reluctance to report wellbeing concerns necessitate proxy measures for identifying support needs. However, discrepancies between parent proxy reports and adolescents' self-reports warrant further research to establish reliable use in referrals. Additional investigations are needed to determine characteristics of a valid proxy reporter, focusing on relationship factors such as closeness, warmth, communication, and time spent together. Moreover, teacher-adolescent concordance research is lacking and warrants attention. This systematic review failed to identify sufficient studies to conduct a meta-analysis. In addition to more studies on teacher-adolescent concordance, and more detailed investigation of predictor variables including age, we strongly encourage researchers to report more detail in their analyses, allowing the field to develop a stronger body of evidence.

1. Introduction

While navigating the developmental hurdles of adolescence, maintaining moderate to high levels of wellbeing presents a challenge for some teens. Adolescence is a period in which an individual navigates pubertal physiological and biological changes, transitions from primary to high school, and moves from parent-centric to peer-centric socialisation (Patalay & Fitzsimons, 2018). This period is considered a pivotal

time for prevention (Kehoe et al., 2014; Lerner et al., 2011; Mueller et al., 2011), with many health and wellbeing programs targeted for this developmental phase (parent and adolescent or teacher and adolescent); Buenconsejo & Datu, 2022; Dariotis et al., 2023; Tejada-Gallardo et al., 2020; Waters, 2011). While adolescents will experience age-related challenges differently (Cummins, 2014; McDougall & Hymel, 2007; Patalay & Fitzsimons, 2016; Tomin & Cummins, 2011), identifying those with poor wellbeing is paramount to ensure they are each given

* Corresponding author.

E-mail addresses: kinorthfield@csu.edu.au (K. Northfield), ASaliba@csu.edu.au (A.J. Saliba), knh.psy@gmail.com (K.M. Harris).

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every opportunity to thrive.

Adolescent wellbeing is defined as an adolescent's positive state of mind and judgement of their own life experience (Cummins et al., 2010). It is commonly evaluated by adults (Achenbach et al., 1987) observing behavioural or social indicators such as sleep patterns, mood, and communication (Headspace, 2020). With poor wellbeing in adolescence linked with adolescent suicide (Sisask et al., 2008), delinquent behaviours (Australian Institute of Health and Welfare, 2011; Suldo & Huebner, 2004), and mental health concerns (Deighton et al., 2014; Kent et al., 2017; Vaz et al., 2014), the need for valid, accessible, and cost-effective measurement tools is unquestionable.

1.1. Measuring adolescent wellbeing

Self-report measures are the most reliable wellbeing assessment tool currently available (for extended review see Cho & Yu, 2020; Kern et al., 2015; Wesley et al., 2020). However, for convenience and practicality, there are situations in which parents, teachers, and other relevant adults act as proxies and make informal judgements to guide support and service decisions (De Los Reyes & Kazdin, 2005; Janicke et al., 2001; Matza et al., 2004). Adolescents most in need of support are unlikely to seek the appropriate support independently (Aguirre Velasco et al., 2020; Nearchou et al., 2018; Rickwood et al., 2005). Concurrently, formal programs to assess wellbeing require extensive resources and would be limited to the data collection points. Employing informal proxy parent or teacher assessments offers convenient, real-time measures.

Proxy assessment in this review refers to a parent (or other caregiver) or teacher completing a wellbeing survey on behalf of the adolescent. They are commonly employed and generally accepted in healthcare settings, where parents or medical practitioners make assessments and decisions on behalf of their family members or patients (Janicke et al., 2001; Matza et al., 2005). The reliability or validity of a proxy's assessment is based on how closely it matches the self-report of the target adolescent (Felder-Puig et al., 2008). Where the proxy and the target adolescents' assessments match, the proxy response would be considered a valid judgement of the adolescents' wellbeing. The level of agreement between the proxy and the target adolescent's own assessment is termed concordance. Given that wellbeing is defined as a subjective judgement, an adolescent's own report is considered correct (Felder-Puig et al., 2008), and deviations from this report are quantified to define levels of agreement or disagreement.

1.2. Concordance noted within the extant literature

Eiser and Morse (2001) demonstrated in a systematic review of concordance studies that included healthy and clinical children up to the age of 18, that agreement between an adolescent's or children's self-report and parents' proxy reports of QOL varied. In comparing clinical and non-clinical populations, they demonstrated that for clinical populations, parents often report their sick children having lower QOL levels than the child's self-report (Eiser & Morse, 2001; Upton et al., 2008). In contrast, parents often reported their healthy children to have higher levels of wellbeing than what the child self-reported (Eiser & Morse, 2001). The degree of concordance between parent proxy and child self-report varied from poor to good across and within studies, making comparisons difficult.

In two previous literature reviews, the influence that child's age and child and parent gender have on concordance of wellbeing reports has been tested, demonstrating inconsistent results (Eiser & Morse, 2001; Upton et al., 2008). Following these and other reviews, researchers are drawing broader conclusions from the concordance data, suggesting that disagreement is not an indication that either reporter is right or wrong; rather, that both reporters offered different information in answering the same question (Upton et al., 2008). Previous reviews of parent-child concordance studies have included a broad age range of target children,

up to 18 years old, and commonly included both clinical and non-clinical adolescents. These two variables are confounding factors that likely increase the variability of the outcomes and have been shown in the extant literature to be predictor variables of wellbeing and concordance.

1.3. Rationale/Aims/Objectives

To date, no systematic reviews focusing on the employment of proxy judgements for healthy adolescents' populations where no prior illness, impairment or special concern have been identified. Systematic reviews that have assessed the concordance between adolescent's or children's and parents' judgement of adolescent wellbeing, have generally included a broad range of participants, including children with clinical or mental health diagnosis and a broad range age range. While parents and teachers drive support and intervention services for adolescents, and adolescents are not independently seeking the necessary support, a measurement tool other than self-report surveys is needed to identify adolescents with poor wellbeing. A proxy assessment may not be the objective measure needed; it appears to be the best available source of current information. As such, the current review questions how well parents and teachers judge the wellbeing of non-clinical adolescents. In addition, we aim to identify the characteristics of a more accurate proxy assessor by interrogating the influence predictor variables such as age and gender have on concordance within the research.

2. Methodology

2.1. Procedure

The protocol for this review was based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA; Page et al., 2021) and was agreed upon with the research team. To identify potentially relevant papers, keyword, abstract, subject and title searches were completed in the SCOPUS, PsycInfo (Ovid), Medline (Ovid), and Ebscohost (all) databases. Search terms used were wellbeing, wellbeing, quality of life, life satisfaction, satisfaction with life, happiness, or positive affect, young, youth, children, teen, teenager, adolescent, student, pupil, minor, underage person, child, or kid, proxy or proxies or informant or cross-informant or inter-rater or interrater or parent rater or teacher rater. MESH (medical subject headings) extensions were included in searches where possible.

2.2. Inclusion criteria

To be included, papers needed to measure adolescent wellbeing using both an adolescent self-report and a parallel parent or teacher proxy report of the target adolescent's wellbeing, and report concordance/agreement or discordance/disagreement between raters. The age criteria limited studies to include participants aged 11–18 years. Where a paper included participants that fell outside of this range, inclusion was considered acceptable if more than 50 % of the participants were within the defined range, and an analysis of agreement between the young person and proxy was reported separately for adolescents. This report focused on assessing healthy populations where no prior illness, impairment, or special concern had been identified.

2.3. Data extraction and synthesis

Endnote (referencing software) was used to store, view, and manage titles, abstracts, and full papers. NVIVO 12, was employed to electronically store, organise, and review content across papers (Saldaña, 2021), and papers were grouped by the measurement tool employed. Consideration of concordance in each domain of wellbeing was noted where possible, and comparisons made. For studies with sufficient information, Fisher's Z scores (FTZ Calculator) were calculated to determine if

correlations differed statistically.

3. Results

3.1. Search results

After duplications were removed, 3094 papers were identified as potentially relevant for this review. Commentary, review papers, unpublished research, or commissioned reports were excluded. The PRISMA diagram (Fig. 1) summarises the search results, screening, and exclusion of studies (Page et al., 2021). Twenty-one papers meeting the criteria were included.

Publication dates ranged from 2003 to 2020, with 80 % of the selected articles being published since 2010. Only one of the selected articles employed a longitudinal study design; the remainder were cross-sectional studies that obtained reports from the adolescent and the parent/teacher close in time. Study samples were commonly obtained in school settings (14 of 21 studies); however, teacher proxies were only featured in two of the 21 selected studies, with parents or other caregivers filling the role as the proxy in all remaining studies. Six of the 21 papers included reports from two proxies, two reported one parent and one teacher report, and four reported judgements from both the mother and father. The quality of the included studies was considered high, all

being judged as satisfactory on six or more of the eight criteria within JBI Critical Appraisal Checklist for Analytical Cross-Sectional Studies (Moola et al., 2020; See appendix B).

While the focus of this review is on adolescent wellbeing, the key terms of QOL and life satisfaction were included to capture reports that employ broader definitions of wellbeing or that have used these terms synonymously. Within the current selection of studies, eight studies employed PedsQL 4.0, four studies employed a version of KIDSCREEN, and four studies utilised KINDL or KINDLR. The remaining studies used various other measures, which touch on constructs that are referred to as wellbeing, some of which have been validated as mental health screening tools (Deighton et al., 2014). PedsQL 4.0, KIDSCREEN and KINDL were each created to measure health-related quality of life (HRQOL); however, in some instances, the term wellbeing defines their measurable domains.

3.2. Summary of concordance

Ten studies reported intra-class correlations (ICC) to define agreement between the two raters, six papers reported Pearson's product moment correlation, one paper reported Spearman's correlation, one reported Mono-trait multi-method correlation, and one reported agreement based on PABAK Kappa. Each of these papers reported

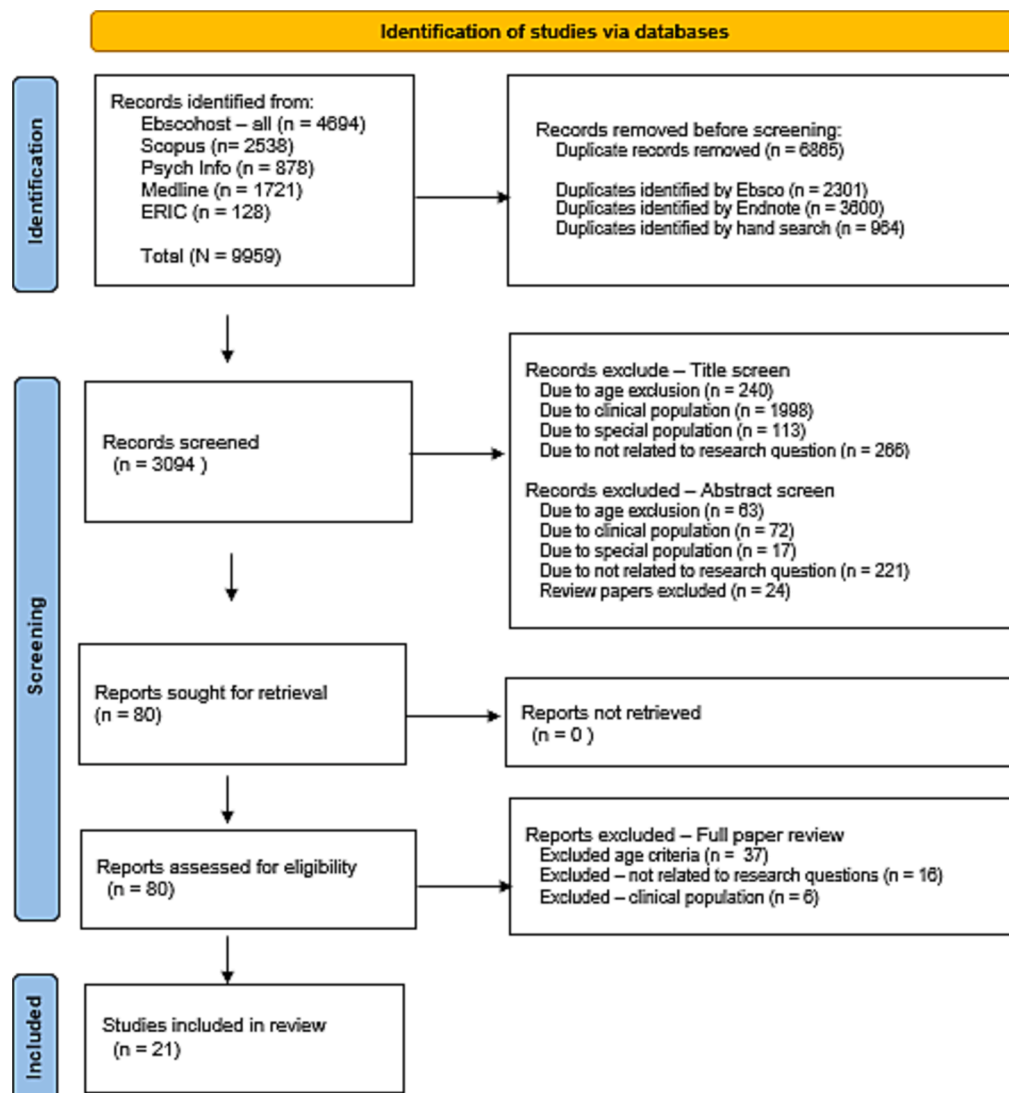


Fig. 1. Prisma Diagram Outlining Study Selection.

agreement or association for each domain, with reasonable variance in agreement levels across domains (see Table 1). Applying the widely accepted interpretation of agreement with ICC (Poor agreement < 0.50, Moderate = 0.50 to 0.75, Good = 0.75 to 0.90, Excellent > 0.90; Koo & Li, 2016) and association with correlation coefficients (small association $r = 0.10$ to 0.30 , medium $r = 0.30$ to 0.50 , large $r > 0.60$; Cohen, 1988), five studies demonstrated good or large association or agreement in one or more domains, with 12 reports indicating moderate agreement or association in at least one domain, and two studies noting only poor agreement or association. Two studies did not report a concordance statistic, focusing on measurement of difference. A comparison of domains with greatest and least agreement did not illicit any clear pattern.

3.3. Influence of adolescent age on concordance between proxy and self-report of adolescent wellbeing

Within this collection of studies, twelve papers (see Table 2) investigated the influence that adolescent age has on concordance between the parent proxy and adolescents' self-reported wellbeing. Five papers reported that age had minimal or no influence on concordance (Bibou-Nakou et al., 2019; Østbye et al., 2010; Ravens-Sieberer et al., 2010; Roeser et al., 2012; Waters et al., 2003). Three papers demonstrated that concordance increases with age (Ellert et al., 2011; Robitail et al., 2007; Seung Hee et al., 2008). Three papers reported a reduction in concordance (Agnihotri et al., 2010; Gandhi et al., 2015; Rajmil et al., 2013),

and one paper demonstrated mixed results (Berkas et al., 2019). Berman et al. (2016), Erhart et al. (2009), and Jozefiak et al. (2008) did not report the influence of age on concordance; however, they noted the adolescent report of wellbeing increased with age, while noting the same pattern had not appeared in the parent report.

3.4. Influence of adolescent gender on concordance between proxy and self-report of adolescent wellbeing

The influence that adolescent gender has on concordance in parent proxy and adolescent self-reports is unclear based on the findings from nine papers that assessed this predictor (see Table 3). Three studies indicated greater agreement exists between parents and daughters (Ellert et al., 2011; Gugliandolo et al., 2015; Robitail et al., 2007). Two studies noted higher agreement between parents and male adolescents (Roeser et al., 2012; Waters et al., 2003), and four demonstrated no difference in agreement due to adolescent gender (Doostfatemehe et al., 2020; Jozefiak et al., 2008; Østbye et al., 2010; Rajmil et al., 2013).

3.5. Maternal or paternal proxy reporter, who is a more accurate proxy reporter of adolescent wellbeing?

Three studies included two parent proxy reports, mother, and father (Gugliandolo et al., 2015; Jozefiak et al., 2008); however, four key papers assessed the influence parent gender has on concordance between

Table 1
Summary of Concordance Outcomes for Studies Included in Review.

Author, year, location	Statistic	Range of concordance	Interpretation of concordance	N	Measure	Domains measured
Waters et al., 2003, Australia	ICC	0.42 to 0.64	Poor to moderate	2096	CHQ PF50 / CHQ CF80	Physical functioning social roles, emotional and mental health and wellbeing, general health, body pain, family functioning
Berman et al., 2016, Sweden	PABAK (item level)	0.18 to 0.82	Slight to substantial	175	KIDSCREEN	Physical wellbeing, psychological wellbeing, autonomy and parents, peers and social support, school environment
Rajmil et al., 2013, Spain	ICC	0.26 to 0.52	Poor to moderate	118		
Ravens-Sieberer et al., 2010, 10 EU countries	r	0.27 to 0.72	Small to large	16,237		
Robitail et al., 2006, 7 EU countries	ICC	0.33 to 0.53	Poor to moderate	1472		
Jozefiak et al., 2008, Norway	r	0.26 to 0.39	Small to medium	586	KINDL or KINDLR	Physical wellbeing, emotional wellbeing, self-esteem, family, friends, school
Roeser et al., 2012, Germany	r	0.55 to 0.77	Large	92		
Ellert et al., 2011, Germany	ICC	0.24 to 0.45	Poor	3372		
Erhart et al., 2009, Germany	ICC	0.24 to 0.45	Poor	4598		
Amiri et al., 2010, Iran	ICC	0.69 to 0.82	Moderate to good	822	PedsQL 4.0	Physical health, psychosocial health summary, emotional functioning scale, social functioning scale, school functioning scale
Berkas et al., 2019, Hungary	ICC	0.56 to 0.58	Moderate	641		
Doostfatemehe et al., 2020, Iran	ICC	0.17 to 0.60	Poor to moderate	573		
Gandhi et al., 2015, USA	ICC	0.40 to 0.51	Poor to moderate	323		
Kaartina et al., 2015, Malaysia	diff	NA	NA	218		
Østbye et al., 2010, Singapore	diff	NA	NA	948		
Seung Hee et al., 2008, Korea	ICC	0.50 to 0.61	Moderate	792		
Reinfjell et al., 2006, Norway	MTMM	0.21 to 0.41	Small to medium	229		
Bibou-Nakou et al., 2019, Greece	r	0.23 to 0.45	Small to medium	3570	SDQ	Emotional problems, peer problems, conduct problems, hyperactivity-inattention prosocial behaviour
Feng and Astell-Burt, 2017, Australia	r	0.47	Medium	3083		Well-being (trait), self-control, emotionality, sociability
Gugliandolo et al., 2015, Italy	r	0.31 to 0.39	Medium	263	TEIQ	Physical, psychological, social relationships, environment
Agnihotri et al., 2010, India	Rho	0.62 to 0.80	Large	515	WHOQOL - BREF	

Note. ICC = Intra-class correlation, Rho = Spearman's correlation, r = Pearson's product moment correlation, MTMM = Mono-trait multi-method correlation, TEIQ = Trait Emotional Intelligence Questionnaire, SDQ = Strengths and Difficulties Questionnaire, CHQ = Child Health Questionnaire.

Table 2
Influence of Adolescent Age as Predictor of Parent/Adolescent Concordance.

Study/Location	Concordance increase or decrease in line with age	Age Range	Measure	How age influence assessed
Agnihotri et al., 2010, India	Decrease	14–16	WHOQOL-BREF	Age-wise correlations
Gandhi et al., 2015, USA	Decrease	15–18	PedsQL 4.0	Regression
Rajmil et al., 2013, Spain	Decrease	13–17	KIDSCREEN	Compare ICC, T1 – vs T2*
Ellert et al., 2011, Germany	Increase	11–17	KINDLR	Compare ICC, 11–13 v 14–17
Robitail et al., 2006, 7 EU countries	Increase	12–18	KIDSCREEN	Regression
Seung et al., 2008, Korea	Increase	13–18	PedsQL 4.0	Identified trend ICC increase with age
Berkes et al., 2019, Hungary	Mixed	13–18	PedsQL 4.0	Compare ICC, 8–12 v 13–18**
Bibou-Nakou et al., 2019, Greece	No change	11–15	SDQ	Comment “no evidence of age-related difference in concordance”
Østbye et al., 2010, Singapore	No change	11–18	PedsQL 4.0	Regression
Ravens-Sieberer et al., 2010, 10 EU countries	No change	12–18	KIDSCREEN	Compare mean t score and effect size 8–11 v 13–18 age group**
Roeser et al., 2012, Germany	No change	11–17	KINDLR	Age covariate
Waters et al., 2003, Australia	No change	12–18	CHQ PF50/CF80	Significance test

Note. * T2 occurred three years after T1 assessment.
** Berkes et al., 2019 and Ravens-Sieberer et al., 2010 employed their broader data set to assess the influence of age.
Regression = Multiple linear regression.

parent proxy and adolescents’ own wellbeing self-report (see Table 4). Jozefiak et al. (2008) and Doostfatemeh et al. (2020) reported minimal or no difference in agreement between paternal and maternal reports. Gugliandolo et al. (2015) reported differences across the four dyad combinations, however when comparing maternal and paternal reports we did not find a significant difference using the Fishers Z calculation. In contrast, Agnihotri et al. (2010) demonstrated that maternal reports were more closely matched to the adolescent self-report. Specifically noting that concordance between mothers and adolescents was greater in psychological and social relations domains, whereas, for the environmental domain, paternal reports had the higher concordance (while not significant – from our calculation) with adolescent self-reports.

3.6. Influence of proxy Reporters’ role as predictor of parent or teacher and adolescent concordance

Four papers in this selection analysed the concordance between multiple proxies: mother and father or parent and teacher combinations. Three of these studies (Feng & Astell-Burt, 2017; Gugliandolo et al., 2015; Jozefiak et al., 2008) found that adult proxy reports (from parents or teachers) exhibited a stronger correlation with each other compared

Table 3
Influence of Adolescent Gender as a Predictor of Parent/Adolescent Concordance.

Study	Concordance increase or decrease predicted by adolescent gender	Measure	How adolescent gender influence assessed
Doostfatemeh, 2020, Iran	No effect	PedsQL 4.0 ^M	DIF analysis
Ellert, 2011, Germany	Daughter greater concordance	KINDLR ^M	Compare of ICC MANOVA *
Gugliandolo et al., 2015, Italy	Daughter greater concordance	TEIQ ^U	MANOVA
Jozefiak, 2018, Norway	No effect	KINDL ^M	Compare correlations. Sign. test difference z scores
Østbye, 2010, Singapore	No effect	PedsQL 4.0 ^M	Gender covariate
Rajmil, 2013, Spain	No effect	KIDSCREEN ^M	GEE
Robitail, 2006, 10 EU countries	Daughter greater concordance	KIDSCREEN ^M	MANOVA Regression
Roeser, 2012, Germany	Son greater agreement	KINDL ^U	Sign. test differences **
Waters, 2003, Australia	Son greater agreement	CHQ PF50/CF80 ^M	Sign. test differences ***

Note. DIF = Differential item functioning, ICC = Intra-class correlation, MANOVA = Multivariate analysis of variance, Regression = Multiple linear regression, GEE = Generalised estimating equation, Sign. test differences = Significance test mean differences.
* Conclusion drawn across multiple domains’ significance of different in global measure of wellbeing not reported.
** For the KINDL-scores the discrepancy between children’s and parents’ ratings was significantly higher in girls (M = – 5.54, SD = 8.30) than in boys [M = – 2.06, SD = 6.33, t(89.22) = – 2.28, p < 0.05].
*** Narrative report gender differences related to discrepancy between parent and adolescent report of mental health.
^M indicates a multidimensional measures vs ^U indicates unidimensional measures were used in comparing the influence of gender on parent / adolescent concordance.

to what was found in any combination of adult and adolescent report. In contrast, one study (Bibou-Nakou et al., 2019) indicated that the strongest association was observed between reports from adolescents and their parents. Bibou-Nakou et al. (2019) and Feng and Astell-Burt (2017) assessed the difference in concordance between one parent proxy and teacher proxy reports both employing the Strengths and Difficulties Questionnaire (SDQ; Goodman & Goodman, 2009). Bibou-Nakou et al. (2019) demonstrated concordance was greatest between parents and adolescents, followed by teachers and adolescents, then parents and teachers. In contrast, Feng and Astell-Burt (2017) found that parents and teachers had the greatest concordance, followed by parents-adolescent, teacher, and adolescent (see Table 5). Going further, they both demonstrated that parents and teachers reported lower difficulty scores for the target adolescent than the adolescent reported for themselves. In each case, the teachers judged the student’s difficulty as being less than what was judged by the parent.

4. Discussion

The key objective of this review was to move beyond quantifying the concordance across reports and identifying the underlying factors that predict greater concordance between adolescent self-report and their proxy report, to determine the characteristics of a more likely accurate proxy assessor. In this systematic review that spans 20 years of research, twenty-one published studies were identified that investigate the use of

Table 4

Influence of Maternal or Paternal Report as Predictor of Parent/Adolescent Concordance.

	Jozefiak et al. (2008)	Doostfatemeh et al. (2020)	Gugliandolo et al. (2015)	Agnihotri et al. (2010)			
	Total scale agreement	Total scale agreement	TEIQ WB	Physical	Psychological	Social relations	Environment
Father/Son	0.26	0.46	0.31	0.76	0.73	0.6	0.81
Father/Daughter	0.34	0.52	0.35	0.72	0.75	0.46	0.80
Mother/Son	0.26	0.50	0.36	0.79	0.81	0.73	0.78
Mother/Daughter	0.39	0.55	0.39	0.68	0.88	0.64	0.80
Fisher's test significance difference: father/son vs mother/son							
Z =	0n.s	−0.878n.s	−0.642n.s	−1.215n.s	−3.204*	−3.806*	1.319n.s
Fisher's test significance difference: father/daughter vs mother/daughter							
Z =	−0.987n.s	−0.71n.s	−0.529n.s	1.269n.s	−6.508*	−4.214*	−0.86n.s

Note. Jozefiak et al. (2008): Age range 8–16 years / KINDL / Pearson's correlation / father-son $n = 589$, Father-daughter $n = 586$, Mother-son $n = 591$, Mother-daughter $n = 589$.

Doostfatemeh et al. (2020): Age range 12–17 years / PedsQL 4.0 / intra-class correlation / mother & father reports for all adolescents $n = 573$.

Gugliandolo et al. (2015): Age range 13–17 years / Trait emotional intelligence questionnaire / Pearson's correlation / mother & father reports for all adolescents $n = 263$.

Agnihotri et al. (2010): Age range 14–16 years / World Health Organization Quality of Life, brief scale / Pearson's correlation mother & father reports for all adolescents $n = 525$.

n.s = $p > 0.05$.

* $p < .05$.

Table 5

Influence of Proxy Reporters' Role as Predictor of Parent or Teacher/Adolescent Concordance Outcome of SDQ Reports.

Feng and Astell-Burt (2017), Australia, 12–13 yrs				
	r – teacher adolescent	r – parent adolescent	r – parent teacher	Fishers Z
Mean Difficulty Score	0.40*	0.47*	0.49*	3.391, $p < .0004$
	N	Mean	SD	
Adolescent report	3083	8.96**	5.52	
Teacher report	3083	5.76**	5.52	
Parent report	3083	7.22**	5.18	
Bibou-Nakou et al. (2019), Greece, 11–15 Yrs				
Subscale Scores	r – teacher adolescent	r – parent adolescent	r – parent teacher	Fisher's Z
Hyperactivity-inattention	0.30	0.45	0.41	−8.004, $p < .0001$
Emotional problems	0.14	0.41	0.16	−13.46, $p < .0001$
Peer problems	0.19	0.39	0.18	−10.028, $p < .0001$
Conduct problems	0.19	0.23	0.19	−1.912, $p = 0.028$
Mean Difficulty Score	N	Mean	SD	
Adolescent report	5477	8.80	5.68	
Parent report	3570	8.10	6.47	
Teacher report	5037	7.39	6.47	

Note. * Correlation between mean difficulty score between reporters.

Mean difficulty score = All sub scores in SDQ except the prosocial score are added to obtain a total difficulties score.

** Feng and Astell-Burt (2017) reported group means and standard deviations; these means have been collapsed into relevant groups according to Cochrane guidelines (Higgins et al., 2019) to ensure accurate means and standard deviations are reported here.

Bibou-Nakou's range of correlations is noted for all subscales except for prosocial, in line with reported means. Means and correlations were not reported for TDS.

Fisher's test of significance evaluates the difference in r between parent/adolescent vs. teacher/adolescent.

SDQ: Strength and difficulty questionnaire (Goodman & Goodman, 2009).

adult proxies in making parallel assessments of adolescent wellbeing. In considering the levels of concordance reported across studies, a reasonable amount of variance is noted in measurement, outcomes, and explanations. This review has identified adolescent age or developmental stage (early, mid, late adolescence), adolescent gender, and parent gender as key factors that may predict a more accurate proxy

assessor. Including teachers as alternate proxy assessors warrants greater investigation given many of these studies are conducted in school settings, with some reports demonstrating that teachers can offer a different perspective in which to make their judgements.

4.1. Adolescent age to explain concordance between proxy and self-report of adolescent wellbeing

The extant literature refers to age as a key predictor of agreement between adolescent and parent proxy reports of adolescent wellbeing (Abate et al., 2018; López-Pérez & Wilson, 2015; Newland et al., 2019). There are two key hypotheses proposed here. First, agreement between parent proxy and adolescent self-report will increase in line with the adolescents' age, due to cognitive maturation (Agnihotri et al., 2010; Jozefiak et al., 2008). Compared to children, adolescents have greater cognitive skills to understand the content and items within the questionnaire and respond appropriately. In addition, adolescents will also have a greater ability to communicate their concerns or issues with their parents, allowing parents greater insight into their troubles, leading to higher levels of agreement. Three of the papers in this collection that assessed the influence of age ($n = 12$) found that agreement increased with age (Ellert et al., 2011; Robitail et al., 2006; Seung Hee et al., 2008). However, the connection to cognitive maturation or increased communication is yet to be tested. Future research could test this hypothesis by including cognitive ability tests or assessing the amount and quality of communication between adolescents and parents as a predictor of concordance.

In this review, three studies (Agnihotri et al., 2010; Gandhi et al., 2015; Rajmil et al., 2013) indicated that the agreement between parents and adolescents' judgement of the adolescent's wellbeing decreased as adolescents grew older. This finding lends support to the alternate age-related hypothesis, suggesting that the decline in agreement stems from developmentally appropriate changes in the adolescent-parent relationship. Adolescents, in their quest to establish their own identity, often create some emotional and physical distance from their families (Peterson, 2014), resulting in reduced emotional openness (Erwin, 1998; López-Pérez & Wilson, 2015), decreased time spent together (Albert et al., 2013), and increased conflict between parents and adolescents (Noller et al., 1992; Van Doorn et al., 2011). Consequently, parents may perceive their adolescents' wellbeing more negatively, and this limited closeness and information shared may impact parents' ability to accurately assess their adolescents' wellbeing. It is important to note that further investigation is required to establish the causal links between these factors. Exploring family connections could provide

valuable insights for identifying a more reliable or accurate proxy assessor. It is important to consider how identity formation in adolescents may differ across cultures before drawing conclusions based on broadly accepted western or individualistic cultural developmental processes (Arnett, 2013; Butten et al., 2021; Patton et al., 2016).

Research indicates that low wellbeing peaks for adolescents during mid adolescence (14–17 years), with higher levels of wellbeing being reported in early and late adolescence (Berman et al., 2016; Casas et al., 2007). This presents two challenges to assessing concordance. First, it is possible that proxies are not aware of this curvilinear pattern of wellbeing in their adolescent and their reports do not reflect this age-related shift. Second, most often, data for adolescents is interrogated as one homogeneous group to determine patterns of concordance. More precise statistical interrogations that account for early, mid, and late adolescence (Berk, 2019; Bluth & Blanton, 2015; De Goede et al., 2009a; Obradovic et al., 2006; Peterson, 2014), may offer different outcomes.

This finding lends support to the alternate age-related hypothesis, suggesting that the decline in agreement stems from developmentally appropriate changes in the adolescent-parent relationship. Adolescents, in their quest to establish their own identity, often create some emotional and physical distance from their families (Noller & Atkin, 2014; Peterson, 2014), resulting in reduced emotional openness (Erwin, 1998; López-Pérez & Wilson, 2015), decreased time spent together, changes to communication patterns (Collins & Laursen, 2009), adjustments to parental monitoring (Dittus et al., 2010; Kerr & Stattin, 2000; Suldo & Huebner, 2004), and increased conflict between parents and adolescents (Noller et al., 1992; Van Doorn et al., 2011). Consequently, parents may perceive their adolescents' emotional state and wellbeing more negatively. Furthermore, this limited closeness and information may impact parents' ability to accurately assess their adolescents' wellbeing. It is important to note that further investigation is required to establish the causal links between these factors. Exploring family connections could provide valuable insights for identifying a more reliable or accurate proxy assessor.

While there is ongoing cognitive maturation, the relationship between adolescents and their parents (family) is appropriately changing. These two competing functions may explain the contradictory and inconclusive findings demonstrated in this review. Mid adolescence is the time in which the parent-adolescent perceived relationship quality is low (McGue et al., 2005), conflict is most frequent (De Goede et al., 2009b; Montemayor, 1990; Van Doorn et al., 2011) and the parental bond is lowest (van Wel et al., 2000), as compared to early and late adolescence. Future research that considers the stage of adolescence (early, mid, late) and assesses curvilinear patterns may lead to more nuanced conclusions that link concordance to familial factors and possibly determine a point in time when proxy assessments are more accurate.

4.2. Adolescent gender to explain concordance between proxy and self-report of adolescent wellbeing

This review indicated a trend of greater concordance in parent proxy and adolescent self-reports of wellbeing between parents and daughters; however, given the variance in unidimensional and multidimensional measures, and reported outcomes, these findings are not conclusive. This trend appears consistent with notions that male children (boys vs. girls) primarily seek autonomy and independence, (Doostfatemeh et al., 2020; Noller & Atkin, 2014) and therefore share less time, and are less close with parents. In addition, girls being defined as more emotionally connected to parents (Branje et al., 2018; Noller & Atkin, 2014; Paulson et al., 1991; Van Lissa et al., 2019), and communicating or confiding more in their parent (Branje et al., 2018; Cava et al., 2014; Izzo et al., 2022; Raffaelli & Duckett, 1989), may offer their parents more information to employ in making their proxy judgements, allowing more accurate proxy reports. While there is support for these possible explanations within the extant literature, they may not adequately address

cultural differences, or variations in family structures (Australian Institute of Health and Welfare, 2011; Blackwell, 2010; Langton & Berger, 2011; Patton et al., 2016). Further research that assesses factors such as closeness and warmth, emotional connection, and communication within the parent/adolescent relationship may better inform what relationship factors predict greater concordance in wellbeing reports. It is important to note that past studies have not examined any possible impact related to gender fluid or non-binary adolescents. This may be an important factor and deserves some attention in future research.

4.3. Maternal reports – Are mothers more accurate proxy reporters of adolescent wellbeing?

The evidence relating to differences in concordance between mothers/adolescents and fathers/adolescents was inconclusive in the current selection of papers. However, while there were no differences between mothers and fathers global report of wellbeing, one study demonstrated that mothers offered a proxy report of adolescent wellbeing that was more closely matched to their adolescent's self-report, specifically in the domains of psychological wellbeing, emotional wellbeing, and social relations. This supports the notion of mothers being more sensitive and having a greater emotional connection with their adolescent children (Doostfatemeh et al., 2015; Gugliandolo et al., 2015; Paulson et al., 1991). Doostfatemeh et al. (2020) hypothesised that daughters might see their mothers as confidants and be more open with them. This is supported by the broader literature showing greater warmth and closeness between adolescents and their mothers (LeCroy, 1988; Thornton et al., 1995; Van Lissa et al., 2019), and connecting adolescent closeness with their mother as a predictor of adolescent wellbeing (Lampropoulou, 2018). These findings may point to concordance levels being influenced by relationship factors. Assessing the predictive value of familial factors such as parental bond, familial conflict, closeness, communication, and shared experiences within parent child relationships may help better identify a more reliable proxy assessor.

4.4. Parents or teachers as proxy reporters – Parents are more accurate

Given the current impetus of school wellbeing programs, it is imperative to investigate how closely teachers' reports of adolescent wellbeing match adolescent self-reports. A large majority of the studies in this review were conducted in school settings; however, only two included reports of adolescent wellbeing as judged by the adolescents' teachers. Bibou-Nakou et al. (2019); Feng and Astell-Burt (2017) assessed the difference in concordance between the parent/adolescent and teacher/adolescent dyad, demonstrating that parent/adolescent reports were more closely matched than teachers/adolescent reports. They further showed that adolescents self-reported the greatest difficulties, followed by their parents, and then teachers. Both studies employed large sample sizes and assessed concordance in outcome scores from the SDQ and evaluated the proxy role (parent or teacher) as a predictor of agreement between the proxy and adolescent. They offer some insight that gives a strong rationale to question further if parents or teachers are more reliable in directing wellbeing support referrals. These reports suggest that parents who may have more exposure to their child's behaviours, emotions, and communication than teachers are more accurate in assessing their challenges. The contrary argument is that adolescents may be less open with parents in discussing their troubles or give less obvious behavioural cues in line with the developmental process of moving away from their family in forming their identity. Further to this, it could be argued that teachers may have a more objective viewpoint due to less emotional ties with their students than parents, and in some cases have undergone training to assess the cues of poor wellbeing. The data here supports the hypothesis that exposure or emotional connection between the parent and adolescent could underlie greater concordance. Further investigations are needed to understand if these same patterns of concordance between parent or

teacher proxies and adolescents would appear with more clearly defined wellbeing measures, as the reference studies are comparing only total difficulties as reported on the SDQ. In addition, including a measure of relationship factors such as warmth, communication, or contact hours may allow researchers to understand better the key variables that influence concordance.

4.5. Wellbeing – The employment of proxy reporters across disciplines

Wellbeing is a construct that crosses many disciplines and by interrogating studies across disciplines, a greater volume of knowledge can be found. This review has identified contradictory positions between disciplines related to the employment of proxy assessors. The importance of the parent proxy is recognised and valued in healthcare settings (Aebi et al., 2017; Killien et al., 2021) with medical intervention and assessments being heavily based on parent and practitioner proxy judgements (Varni et al., 2007). However, within positive psychology, proxy assessments of wellbeing are openly refuted because wellbeing is a personal evaluation or state of mind related to various domains within one's own life (Cummins, 2014). While both positions are theoretically and practically valid, the question relating to the accuracy of the proxy assessors is still unclear. In situations where adolescents are not willing or able to share their wellbeing concerns, parents, teachers, and practitioners remain in dire need of a reliable measurement tool. Anecdotally, there is a presumption that these key stakeholders can offer a judgement that will serve well enough in the absence of an objective measure, and they are being employed to drive services and support. Future research is needed to unpack the differences in position between the two disciplines, with the goal of more closely connecting the research outcomes to inform policy.

4.6. Variance in operational definitions of wellbeing, concordance, and statistical methods employed

A key challenge in evaluating the current selection of research papers has been how wellbeing is operationally defined. While the measures employed in this collection are diverse, they each touch on aspects closely related to wellbeing and the term wellbeing is often noted interchangeably for related constructs within the measures themselves or within their research narrative. However, the purpose of this paper is to assess the concordance between reporters, and the data interrogated here is considered the most appropriate available to do so. While caution in drawing finite conclusions is applied, the data within these studies serve to inform future research to understand better how, when, and who can offer appropriate proxy assessments of adolescents' wellbeing to make service and support referrals.

The current set of studies employed various statistical methods, which may underly some of the variance in outcomes. For example, Doostfatemeht et al. (2020) assessed the influence of adolescent gender on degrees of concordance employing the DIF method, which demonstrates how items function differently for adolescents and their parents, adolescent gender did not affect concordance. The same data analysed with ICC and measuring association between reports on domain or total scale level indicated that greater agreement between girls (over boys) and their parents. There is an ongoing debate to determine the most appropriate way to measure inter-rater agreement (Treveltham, 2016); however, a gold standard is yet to be determined and would be a key factor in advancing this research area.

4.7. Limitations

Comparing concordance across this collection of studies was challenging due to the variety of statistical methods employed to assess agreement between reporters. While a determination of the best statistical method to employ is beyond the scope of this paper, it may be unreasonable to expect that different reporters, who bring different

experience and understanding of the survey items to offer the same response. As noted by (Eiser & Jenney, 2007), each reporter adds a different perspective and supplies different information. Further investigation is warranted and underway to determine how best to compare and employ this different information in clinical, referral, and research settings (Mpundu-Kaambwa et al., 2022).

Most of the studies included in this collection were conducted in school settings, included a measurement of school functioning, and agreement and participation by parents was necessary. These factors limit the study samples to in-school adolescents with some degree of family connection. Out of school adolescents are an important reference group at greater risk of wellbeing concern (Gubbels et al., 2019), as are students living with one or no parents (Berman et al., 2016). Excluding these groups from the research pool is a shortcoming of the wider research area.

Due to the variation in how wellbeing is defined, measured and how concordance has been assessed across this collection of studies, a meta-analysis was not possible. Future reviews could benefit from data being openly available to allow greater scope for reviewers to collate the findings more accurately.

Numerous studies within this review have focussed on gender, role of the parent (mother or father) and adolescent age as potential predictors of concordance. While these are relevant considerations, their application to family systems may be culturally limited, and not be generalizable to collectivist cultures where the transitions, in particular related to identity formations differ to western individualistic cultures often referred to in this paper and more broadly the extant literature (Arnett, 2013; Koepke & Denissen, 2012; Patton et al., 2016).

5. Conclusion

This review highlights the role of proxy reporters in assessing the wellbeing of adolescents and their limitations in comparison to self-reports. It challenges the assumption that parents and other adults can accurately evaluate adolescents' wellbeing for service referrals, suggesting that proxy reporters should be viewed as supplementary sources of information. Furthermore, the review aimed to identify characteristics that contribute to a more accurate proxy assessor. Findings from various studies showed mixed results regarding the predictive value of adolescent age and gender, and parents' role.

The influence of adolescent age on the concordance between parent and adolescent reports of adolescent wellbeing remains inconclusive. Given the fluctuations in wellbeing and shifts in familial relationships during this developmental stage, it is imperative for future studies to investigate potential curvilinear patterns of concordance. Specifically, it is reasonable to anticipate reduced agreement between parent proxy reports and adolescents' self-reports during mid-adolescence, a period characterized by heightened familial conflict and diminished adolescent wellbeing. A better understanding of how age or stage of adolescence affects concordance could aid in determining when a proxy report is more likely to offer a valid indication of the adolescent's wellbeing.

While teachers are often referral agents for adolescent wellbeing support services, evaluation of their judgements has been broadly overlooked in the literature. This review highlights the necessity of further investigation to evaluate teacher proxy reports of adolescent wellbeing, and further validate their use in referral pathways for student support services.

Patterns emerged that may indicate that mothers' reports align more closely with their adolescents' self-reports, in psychological and emotional domains, and daughters' reports may match more closely with parents' proxy judgements. Notably, the quality of the adolescent/proxy relationship has been inadequately assessed in concordance studies. Factors such as closeness, warmth, emotional connection, communication, and shared time play a crucial role in defining this relationship and influencing the quantity and quality of information available to proxies for making judgments. These relationship factors are

likely to predict a more valid proxy assessment and potentially identify a more reliable proxy assessor of adolescent wellbeing.

CRedit authorship contribution statement

Kirstie Northfield: Conceptualization, Methodology, Data curation, Writing – original draft, Writing – review & editing. **Anthony Saliba:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Doctor Keith Harris:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. PRISMA 2020 Checklist

Checklist for Abstract.

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	No
Synthesis of results	6	Specify the methods used to present and synthesise results.	No
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	No discussion section in Abstract
Interpretation	10	Provide a general interpretation of the results and important implications.	
OTHER			
Funding	11	Specify the primary source of funding for the review.	NA
Registration	12	Provide the register name and registration number.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. <https://doi.org/10.1136/bmj.n71>.

For more information, visit: <https://www.prisma-statement.org/>.

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Yes
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Yes
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Yes
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Yes
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Yes
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Yes
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Yes
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Yes
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Yes

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Section and Topic	Item #	Checklist item	Location where item is reported
Study risk of bias assessment	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Yes
	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Yes
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Yes
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Yes
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Yes
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Yes
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If <i>meta</i> -analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Yes
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, <i>meta</i> -regression).	Yes
Reporting bias assessment	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Yes
	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Yes
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Yes
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Yes
Study characteristics	17	Cite each included study and present its characteristics.	Yes
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Yes – appendix
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Yes
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Yes
	20b	Present results of all statistical syntheses conducted. If <i>meta</i> -analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Yes
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Yes
Reporting biases	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Yes
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Yes
	23b	Discuss any limitations of the evidence included in the review.	Yes
	23c	Discuss any limitations of the review processes used.	Yes
	23d	Discuss implications of the results for practice, policy, and future research.	Yes
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	NA
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	NA
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	NA
Declaration of Competing Interests	26	Declare any competing interests of review authors.	NA
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

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For more information, visit: <https://www.prisma-statement.org/>.

Appendix B. . JBI Critical Appraisal Checklist for analytic Cross-Sectional studies

	Were the criteria for inclusion in the sample	Were the study subjects and the setting described in detail	Was the exposure measured in a valid and reliable way	Were objective, standard criteria used for measurement	Were confounding factors identified	Were strategies to deal with the confounding factors stated	Were the outcomes measured in a valid and reliable way	Was appropriate statistical analysis used
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			Were the criteria for inclusion in the sample clearly defined	Were the study subjects and the setting described in detail	Was the exposure measured in a valid and reliable way	Were objective, standard criteria used for measurement of the condition	Were confounding factors identified	Were strategies to deal with the confounding factors stated	Were the outcomes measured in a valid and reliable way	Was appropriate statistical analysis used		
			clearly defined			of the condition						
#	Author	Year	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Score	Include
1	Agnihotri	2010	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
2	Amiri	2010	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
3	Berkes	2019	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
4	Berman	2016	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
5	Bibou-Nakou	2019	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
6	Doostfatemeh	2020	✓	✓	✓	✓	✓	✓	✓	✓	7	YES
7	Ellert	2011	✓	✓	✓	✓	✓	×	✓	✓	8	YES
8	Erhart	2009	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
9	Feng	2017	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
10	Gandhi	2015	✓	✓	✓	✓	✓	×	✓	✓	7	YES
11	Gugliandolo	2015	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
12	Jozefiak	2008	✓	✓	✓	✓	✓	×	×	✓	6	YES
13	Kaartina	2015	✓	✓	✓	✓	✓	×	×	✓	6	YES
14	Ostbye	2010	✓	✓	✓	✓	✓	×	✓	✓	7	YES
15	Rajmil	2013	✓	✓	✓	✓	✓	✓	✓	✓	7	YES
16	Ravens- Sieberer	2010	✓	✓	✓	✓	✓	×	✓	✓	7	YES
17	Reinfjell	2006	✓	✓	✓	✓	✓	✓	✓	✓	8	YES
18	Robitail	2006	✓	✓	✓	✓	✓	×	✓	✓	7	YES
19	Roeser	2012	✓	✓	✓	✓	✓	×	✓	✓	7	YES
20	Seung	2008	✓	✓	✓	✓	✓	×	✓	✓	7	YES
21	Waters	2003	✓	✓	✓	✓	×	×	✓	✓	6	YES

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